

GOSFORD HIGH SCHOOL
Mathematics Assessment Task No 1
December 2006

- Time allowed - 60 minutes plus 5 minutes reading time
- All necessary working must be shown
- Full marks may not be awarded for untidy work or work that is poorly set out
- Begin each new question on a new page

Question 1

- a) The quadratic equation $x^2 - 3x - 5 = 0$ has roots α and β . What is the value of:
- i) $\alpha + \beta$ (1)
 - ii) $\alpha\beta$ (1)
 - iii) $\alpha^2 + \beta^2$ (2)
- b) A quadratic equation has roots α and β . Write down the equation if $\alpha + \beta = 3$ and $\alpha\beta = -1$. (2)
- c) Find the value of k in $x^2 + 6x + k = 0$ if one root is double the other. (2)
- d) Find the values of k for which $x^2 + 6x - 3k = 0$ has real roots. (2)
- e) Find the minimum value of $2x^2 - 12x + 7$. (2)

Question 2

- a) Solve $3x^2 - 5x + 1 = 0$ expressing answers in simplest surd form. (2)
- b) Solve $x^4 - 10x^2 + 9 = 0$ (3)
- c) Differentiate
- i) $x\sqrt{x}$ (1)
 - ii) $(3x^2 - 7)^{10}$ (1)
 - iii) $\frac{2x - 5}{3x + 7}$ (2)
 - iv) $2x(3 - x)^4$ (2)

Question 3

- a) Find exact values for p if $px^2 - 2x + 3p = 0$ is negative definite. (3)
- b) Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$ (2)
- c) The curve $y = x^3 + ax^2 + 7x - 5$ has a stationary point when $x = 1$. Find the value of a . (2)
- d) Find $\frac{d^2y}{dx^2}$ if $y = 3x^4 - x^3$ (2)
- e) A continuous curve $y = f(x)$ has the following properties for the interval $a \leq x \leq b$:
- $$f(x) > 0, \quad f'(x) > 0 \text{ and } f''(x) < 0$$
- Sketch a curve satisfying these conditions and state the greatest value of $f(x)$ in this interval. (3)

Question 4

- a) i) Show that the gradient of the tangent to the curve $y = \frac{1}{4}x^2(x^2 - 7)$ at the point A (2, -3) is 1. Hence find the equation of this tangent. (4)
- ii) Determine the equation of the tangent at the point B (-2, -3) on the curve. (3)
- iii) Show that the angle between the tangents at A and B is 90° and calculate the point of intersection of these tangents. (2)
- b) Find the equation of the normal to the parabola $y = 3 + 6x - 2x^2$ at the point where $x = 1$. (3)

Question 5

- a) Given $f(x) = x^3 - 6x^2 + 9x - 5$, find for what values of x the function is:
- i) Increasing (2)
- ii) concave up (2)
- b) Find any stationary points on this curve and determine their nature. (4)
- c) Neatly sketch the curve for $0 \leq x \leq 5$, clearly labelling all critical points and the y intercept. (3)
- d) What are the greatest and least values of the function in the interval $0 \leq x \leq 5$? (2)

Question 6

- a) For the parabola $y = \frac{1}{6}x^2$, find:
- i) the co-ordinates of the focus (2)
 - ii) the equation of the directrix (1)
 - iii) the length of the latus rectum (2)
- b) i) A and B are the points $(-3, -1)$ and $(7, 3)$ respectively.
The point $P(x, y)$ moves so that $\angle APB = 90^\circ$.
Show that the equation of the locus of P is the circle with equation
 $x^2 - 4x + y^2 - 2y = 24$. (2)
- ii) State the co-ordinates of the centre and the length of the radius of this circle. (2)
- c) Find the co-ordinates of the vertex and the focus of the parabola $2y = (x - 1)^2 - 4$ (2)

2006 - 2u Task 1

1 a) $x^2 - 3x - 5 = 0$

i) $\alpha + \beta = -\frac{(-3)}{1} = 3$

ii) $\alpha\beta = -5$

iii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
 $= (3)^2 - 2(-5)$
 $= 9 + 10$
 $= 19$

b) $\alpha + \beta = 3, \alpha\beta = -1$
 $x^2 - (\alpha + \beta)x + \alpha\beta = 0$
 $\therefore x^2 - 3x - 1 = 0$

c) $x^2 + 6x + k = 0$

let roots be $\alpha, 2\alpha$

$$\begin{array}{l|l} \therefore \alpha + 2\alpha = -6 & 2\alpha^2 = k \\ 3\alpha = -6 & \therefore k = 2(-3)^2 \\ \alpha = -3 & = 18 \end{array}$$

d) $x^2 + 6x - 3k = 0$

real roots $\rightarrow \Delta \geq 0$

$$\Delta = 36 + 12k$$

$$\therefore 36 + 12k \geq 0$$

$$12k \geq -36$$

$$k \geq -3$$

e) $2x^2 - 12x + 7$

$$\begin{array}{l|l} \text{vertex at } x = \frac{-b}{2a} & \therefore \text{min. value} \\ = \frac{12}{4} & \text{at } 2(9) - 36 + 7 \\ = 3 & = -11 \end{array}$$

Q2

a) $3x^2 - 5x + 1 = 0$

$$x = \frac{5 \pm \sqrt{25 - 12}}{6}$$
$$= \frac{5 \pm \sqrt{13}}{6}$$

b) $x^4 - 10x^2 + 9 = 0$

let $m = x^2$

solve $m^2 - 10m + 9 = 0$

$$(m-9)(m-1) = 0$$

$$m = 9$$

$$x^2 = 9$$

$$x = \pm 3$$

$$m = 1$$

$$x^2 = 1$$

$$x = \pm 1$$

c) i) $\frac{d}{dx}(x^{3/2}) = \frac{3}{2}x^{1/2}$
 $= \frac{3\sqrt{x}}{2}$

ii) $\frac{d}{dx}(3x^2 - 7)^{10} = 10(3x^2 - 7)^9(6x)$
 $= 60x(3x^2 - 7)^9$

iii) $\frac{d}{dx}\left(\frac{2x-5}{3x+7}\right)$

$$= \frac{(3x+7)(2) - (2x-5)(3)}{(3x+7)^2}$$

$$= \frac{6x+14 - 6x+15}{(3x+7)^2}$$

$$= \frac{29}{(3x+7)^2}$$

$$\text{iv) } \frac{d}{dx} (2x(3-x)^4)$$

$$u = 2x$$

$$v = (3-x)^4$$

$$u' = 2$$

$$v' = 4(3-x)^3(-1)$$

$$= -4(3-x)^3$$

$$= (3-x)^4(2) + (2x)4(3-x)^3$$

$$= 2(3-x)^3[(3-x) - 4x]$$

$$= 2(3-x)^3(3-5x)$$

Q3

$$\text{a) } px^2 - 2x + 3p = 0$$

neg. def $p < 0$, $\Delta < 0$

$$\Delta = 4 - 4(p)(3p)$$

$$= 4 - 12p^2$$

$$\therefore 4 - 12p^2 < 0$$

$$4(1 - 3p^2) < 0$$

$$4(1 - \sqrt{3}p)(1 + \sqrt{3}p) < 0$$

$$\therefore p < 1 - \sqrt{3}$$

$$\begin{array}{c} 1+\sqrt{3} \\ \hline 1-\sqrt{3} \end{array}$$

$$\text{b) } \lim_{x \rightarrow 3} \frac{(x-3)(x+3)}{x-3}$$

$$= 3+3$$

$$= 6$$

$$\text{c) } y = x^3 + ax^2 + 7x - 5$$

$$y' = 0 \text{ at } x = 1$$

$$y' = 3x^2 + 2ax + 7$$

$$\therefore 0 = 3 + 2a + 7$$

$$\therefore 2a = -10$$

$$a = -5$$

$$\text{d) } y = 3x^4 - x^3$$

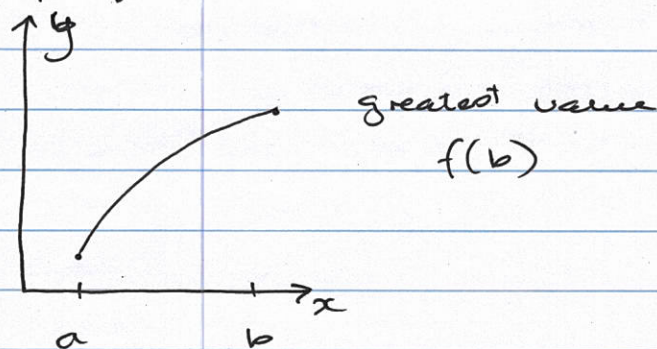
$$y' = 12x^3 - 3x^2$$

$$\frac{d^2y}{dx^2} = 36x^2 - 6x$$

e) $f(x) > 0 \rightarrow$ above x axis

$f'(x) > 0 \rightarrow$ +ve gradient $\rightarrow$ rising

$f''(x) < 0 \rightarrow$ concave down



Q4

$$\text{a) i) } y = \frac{1}{4}x^2(x^2 - 7)$$

$$y = \frac{1}{4}x^4 - \frac{7}{4}x^2$$

$$y' = x^3 - \frac{7}{2}x$$

at $x = 2$

$$m \text{ of tangent} = (2)^3 - \left(\frac{7}{2}\right)(2)$$

$$= 8 - \frac{7}{2} \times 2$$

$$= 8 - 7$$

$$= 1$$

$\therefore$ eqn of tangent

$$y + 3 = 1(x - 2)$$

$$y + 3 = x - 2$$

$$y = x - 5$$

ii) at $x = -2$

$$m \text{ of tangent} = (-2)^3 - \frac{7}{2}(-2)$$

$$= -8 + 7$$

$$= -1$$

∴ eqn of tangent

$$y+3 = -1(x+2)$$

$$y+3 = -x-2$$

$$x+y+5=0$$

iii) m_1 of tangent at A = 1

m_2 of tangent at B = -1

$$\therefore m_1 \times m_2 = 1 \times -1 = -1$$

∴ tangent at A $\perp$ tangent at B

∴ at 90° to each other.

pt. intersection at:

$$x+(x-5)+5=0$$

$$2x=0$$

$$x=0 \quad (0, -5)$$

$$\therefore y = -5$$

b) $y = 3 + 6x - 2x^2$ at $x=1$ normal

$$y' = 6 - 4x$$

$$\text{at } x=1 \quad m \text{ of tangent} = 6 - 4 = 2$$

$$\therefore m \text{ of normal} = -\frac{1}{2}$$

$$x=1, y = 3 + 6 - 2$$

$$y = 7 \quad (1, 7)$$

∴ eqn normal

$$y - 7 = -\frac{1}{2}(x - 1)$$

$$2y - 14 = -x + 1$$

$$x + 2y - 15 = 0$$

Q5

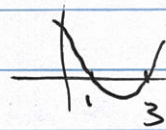
$$a) f(x) = x^3 - 6x^2 + 9x - 5$$

i) increasing $f'(x) > 0$

$$f'(x) = 3x^2 - 12x + 9$$

$$= 3(x^2 - 4x + 3)$$

$$= 3(x-3)(x-1)$$



∴ increasing at $x > 3, x < 1$

ii) concave up $f''(x) > 0$

$$f''(x) = 6x - 12$$

$$\therefore 6x - 12 > 0$$

$$6x > 12$$

$$x > 2$$

b) stat pts at $f'(x) = 0$.

$$3(x-3)(x-1) = 0$$

$$x = 3$$

$$x = 1$$

$$y = -5$$

$$y = -1$$

$$f''(3) = 6 > 0$$

$$f''(1) = -6 < 0$$

∴ minimum

∴ maximum turning

turning pt

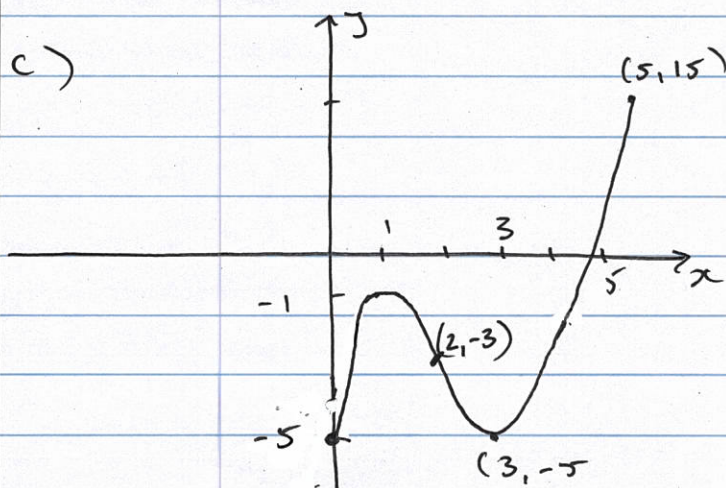
pt

at $x=3$

at $x=1, y=-1$

$$y = -5$$

c)



d) greatest value = 15
least value = -5

Q6

a) $y = \frac{1}{6}x^2$

i) $4a = \frac{1}{6}$
 $a = \frac{1}{24}$

$\therefore$ focus $(0, \frac{1}{24})$

ii) directrix = $y = -\frac{1}{24}$

iii) length of latus rectum = $4 \times \frac{1}{24}$
 $= \frac{1}{6}$

b) i) $A(-3, -1), B(7, 3) \quad P(x, y)$

$m(AP) \perp m(BP)$

$m(AP) = \frac{y+1}{x+3}$

$m(BP) = \frac{y-3}{x-7}$

$\therefore \frac{y+1}{x+3} \times \frac{y-3}{x-7} = -1$

$\therefore y^2 - 3y + y - 3 = -(x^2 - 7x + 3x - 21)$

$y^2 - 2y - 3 = -x^2 + 4x + 21$

$(x^2 - 4x + 4) + (y^2 - 2y + 1) = 21 + 3 + 4$
 $+1$

$(x-2)^2 + (y-1)^2 = 29$

or $x^2 - 4x + y^2 - 2y = 24$

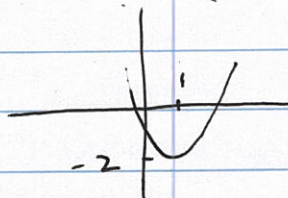
ii) centre $(2, 1)$ radius = $\sqrt{29}$

c) $2y = (x-1)^2 - 4$

$2y + 4 = (x-1)^2$

$2(y+2) = (x-1)^2$

$\therefore$ vertex $(1, -2) \quad a = \frac{1}{2}$



focus $(1, -1\frac{1}{2})$